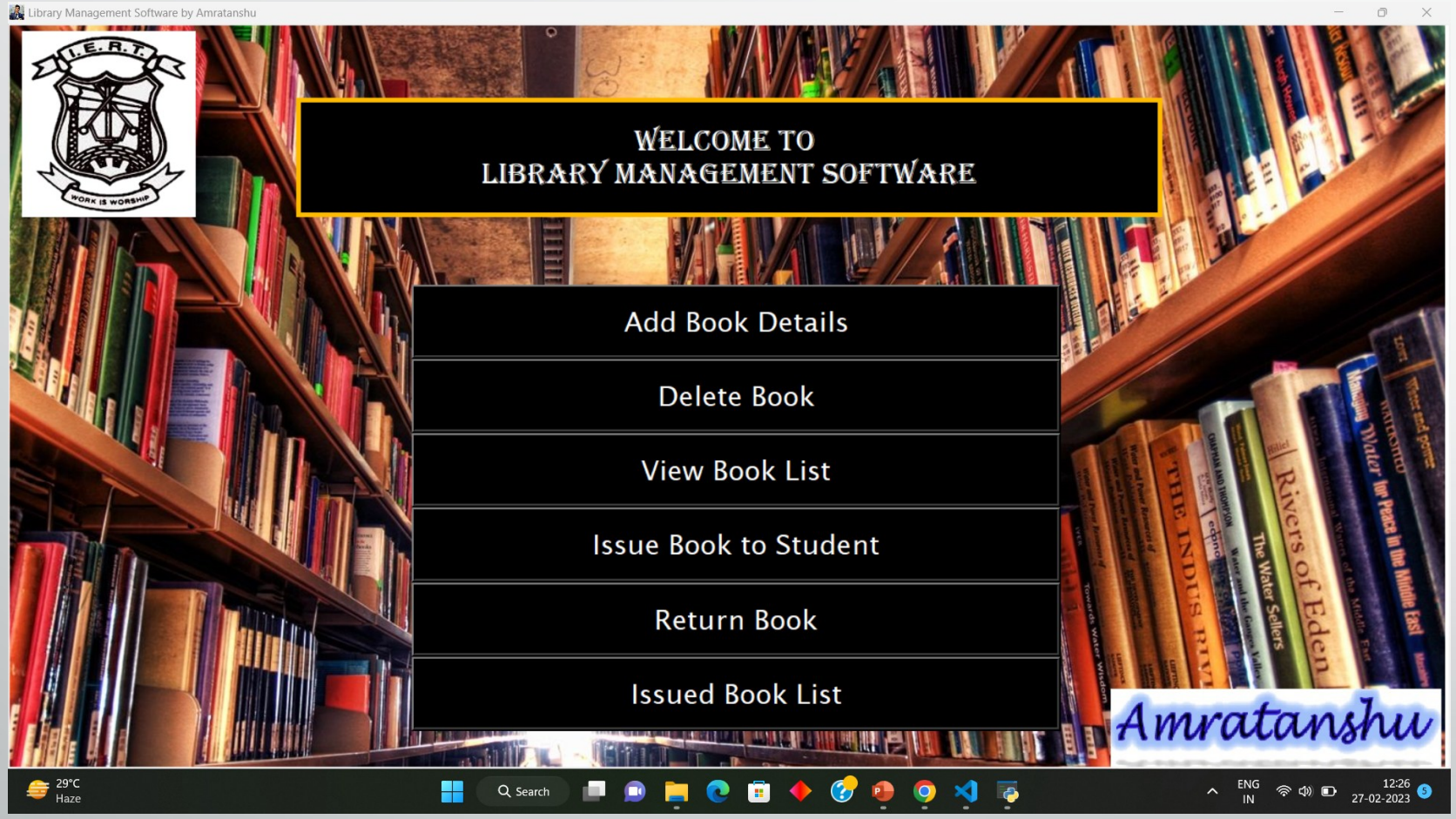
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INTRODUCTION TO PROJECT



The library management deals in management of records. The records include books which are required by different students accordingly. Maintaining these records manually becomes difficult thus we are proving such system a computerized backend.

OBJECTIVES

- To maintain the records in digital manner.
- Reduce the paper work which is hard to maintain.
- Reduce cost as it is one time investment.
- Reduce man-power required, as one person can manage software.
- Promoting Digital India campaign of Government of India

METHODOLOGY USED

- For front-end we have used Python3 which is a high level programming language.
- For GUI we have used 'tkinter' toolkit of Python.
- For back-end i.e., database we have used MySQL which is RDBMS (Relative Database Management System)
- There are six modules in the project:
  1. Add Book Details
  2. Delete Book
  3. View Book List
  4. Issue Book to Student
  5. Return Book
  6. Issued Book List





METHODOLOGY USED

- There are two tables in the back-end in one database
- Structure of Tables is shown:

Field	Type	Null	Key	Default	Extra
bid	varchar(50)	NO	PRI	NULL	
title	varchar(40)	YES		NULL	
author	varchar(40)	YES		NULL	
status	varchar(30)	YES		NULL	

Database name: **db**

Tables_in_db
books
issued_books

Structure of **issued_books** table.

Field	Type	Null	Key	Default	Extra
bid	varchar(50)	NO	PRI	NULL	
issuedto	varchar(40)	YES		NULL	

MODULE DESCRIPTION

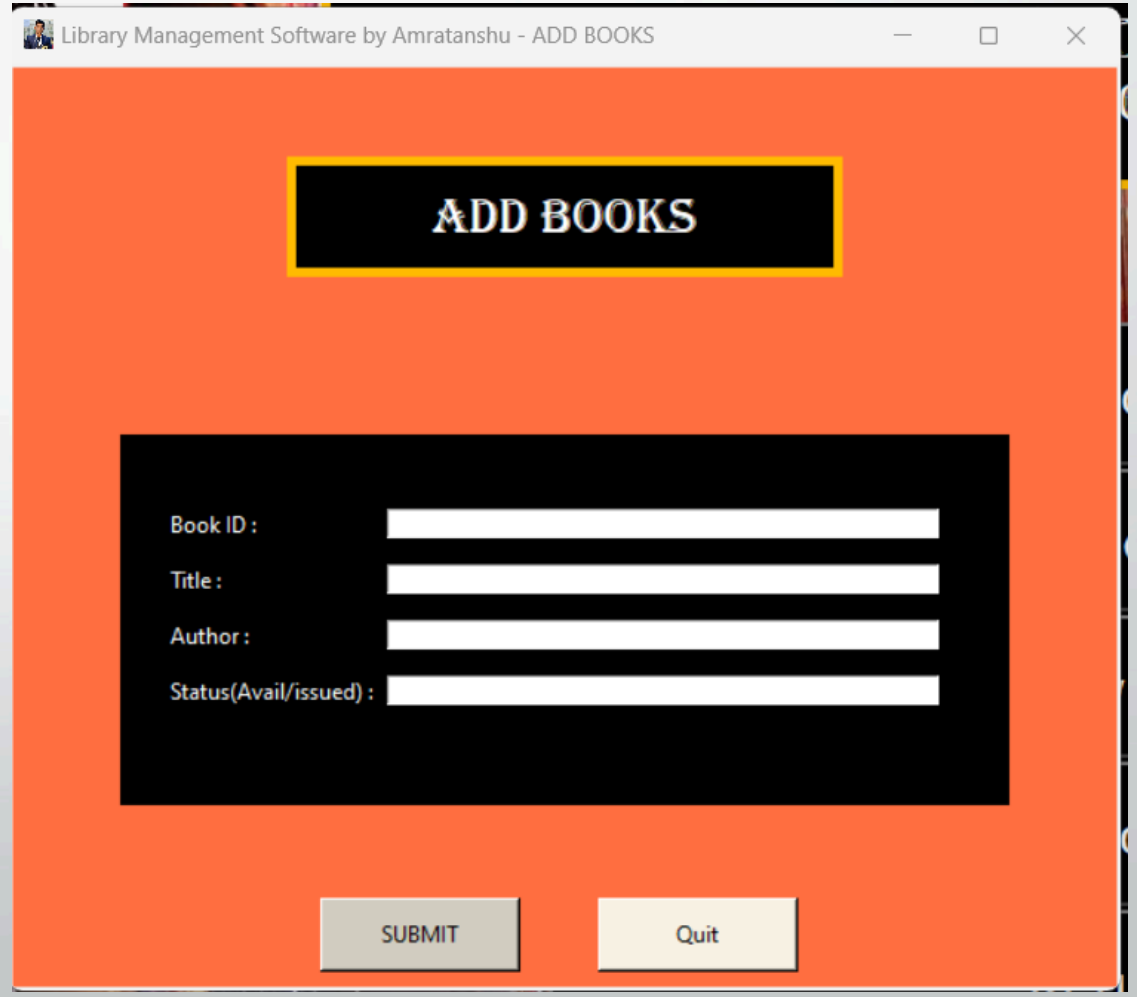
Module: ADD BOOKS

This module adds the books into database in the table **books.**

It takes four input from user and adds them in the books table

The code for providing the connection to back-end is:

```
mypass =  [credential redacted]
mydatabase="db" #The database name
con = pymysql.connect
(host="localhost",user="root",password=mypa
ss,database=mydatabase)
cur = con.cursor() #cur -> cursor
```



MODULE DESCRIPTION

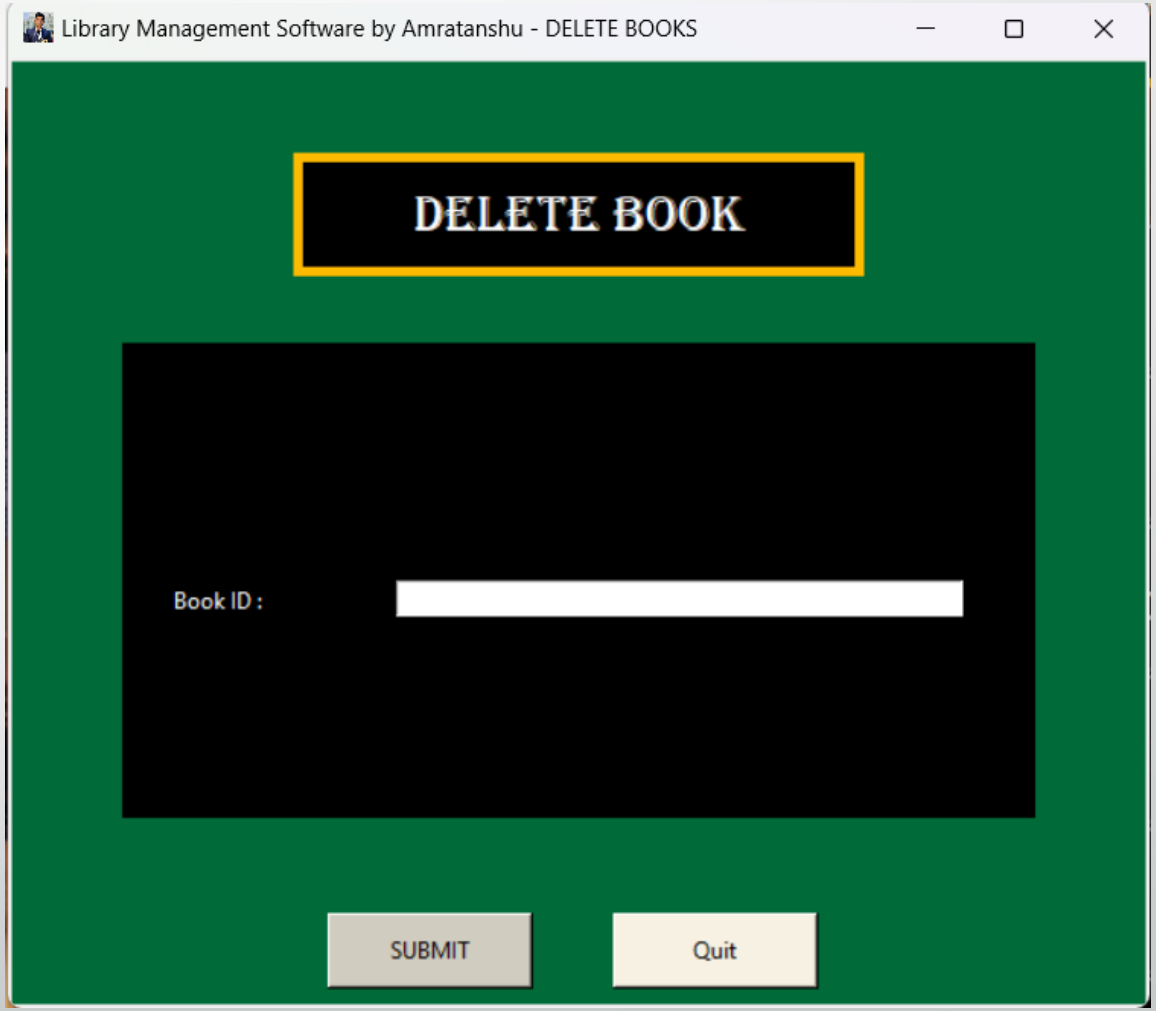
Module: DELETE BOOKS

This module deletes the books from the database from the table **books and issued_books.**

It takes book ID as input from user and removes the data from both the tables.

The query to delete the book is

```
deleteSql = "delete from "+bookTable+"
where bid = '"+bid+"'"
deleteIssue = "delete from
"+issueTable+" where bid = '"+bid+"'"
```



MODULE DESCRIPTION

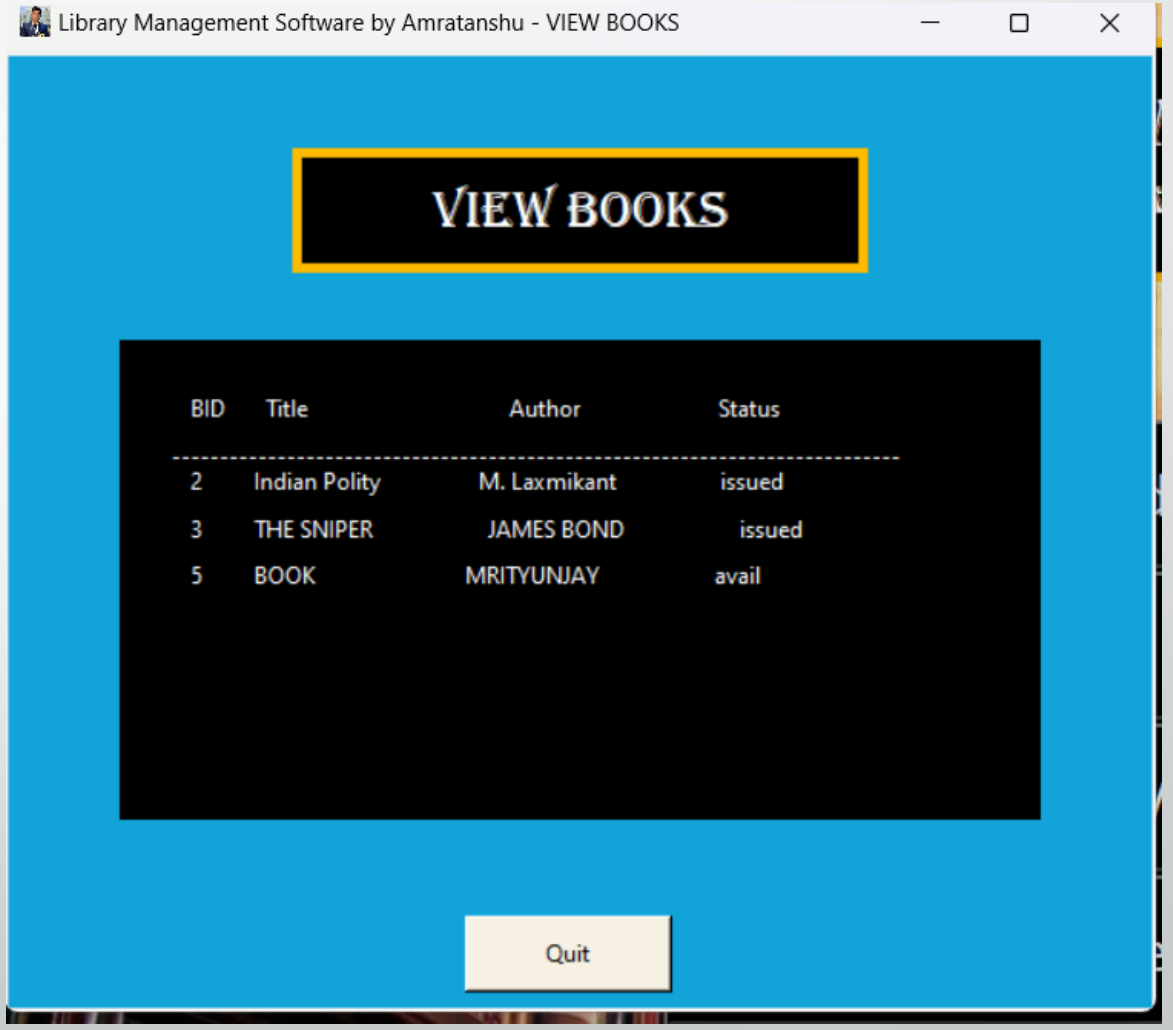
Module: VIEW BOOKS

This module views a table of books from the database from the table **books.**

It does not take any input from user and displays the list from the back-end.

The code to get book list from database is:

```
getBooks = "select *
from "+bookTable
```



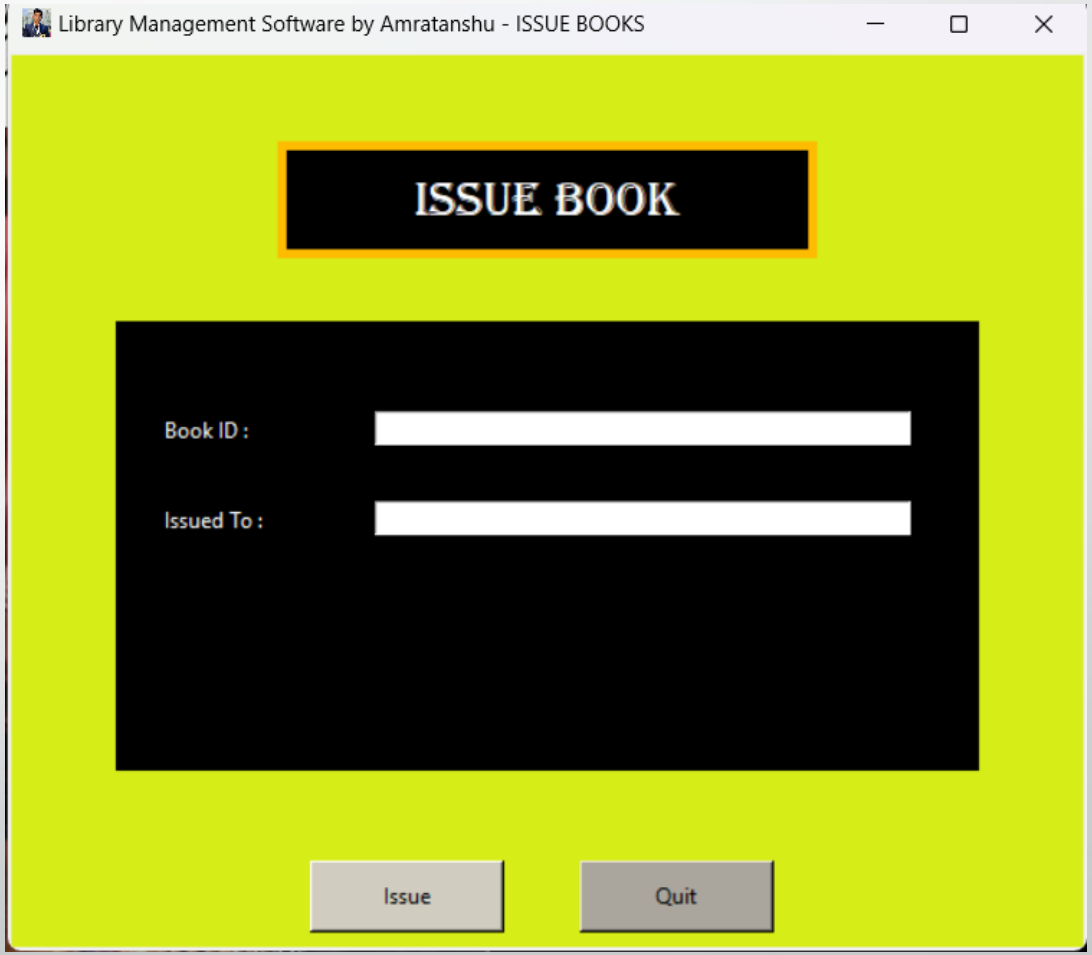
MODULE DESCRIPTION

Module: ISSUE BOOK

This module takes two input from user and adds the data to issued_books table.

The code for issue books is:

```
extractBid = "select bid
from "+bookTable
```



MODULE DESCRIPTION

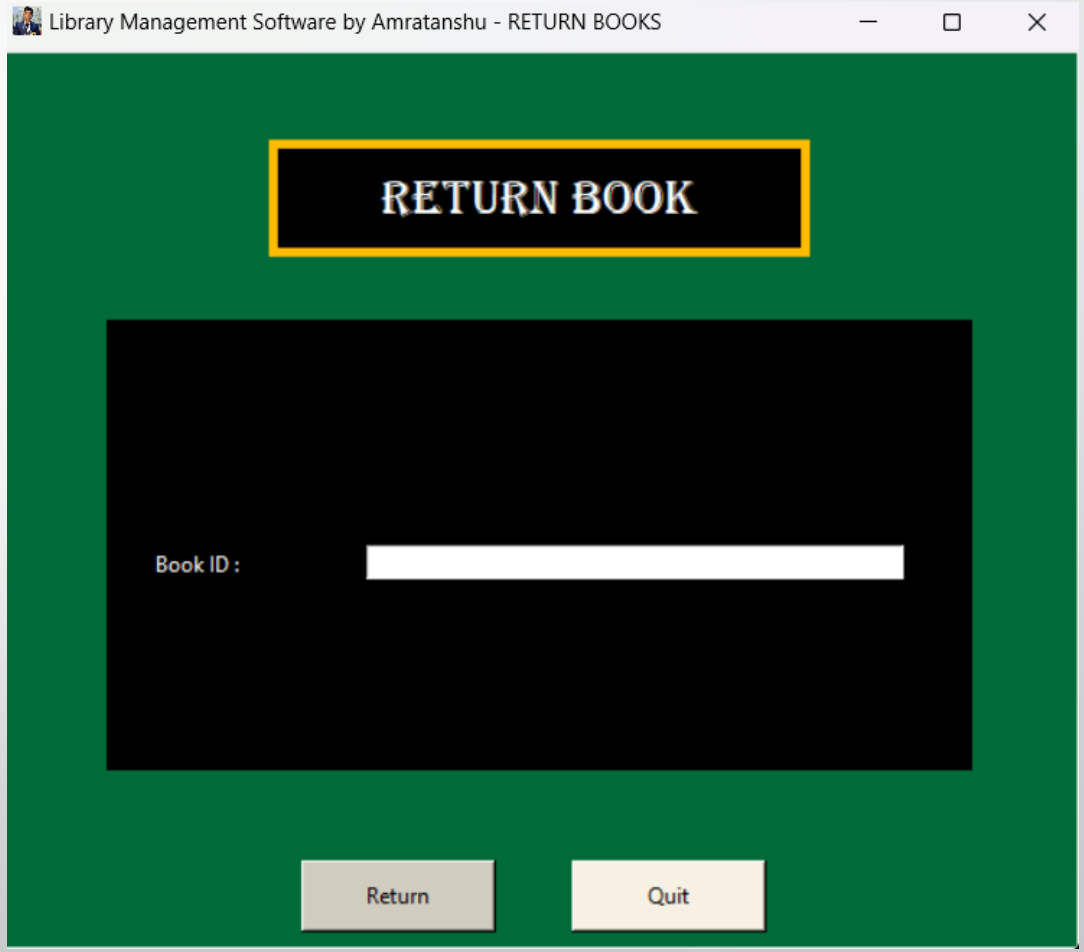
Module: RETURN BOOKS

This module takes book ID as input and removes the name of issuer from **issued_book** table.

It deletes the data from from issued_books table.

The code for Return books is:

```
updateStatus = "update
"+bookTable+" set status =
'avail' where bid = '"+bid+"'"
```



MODULE DESCRIPTION

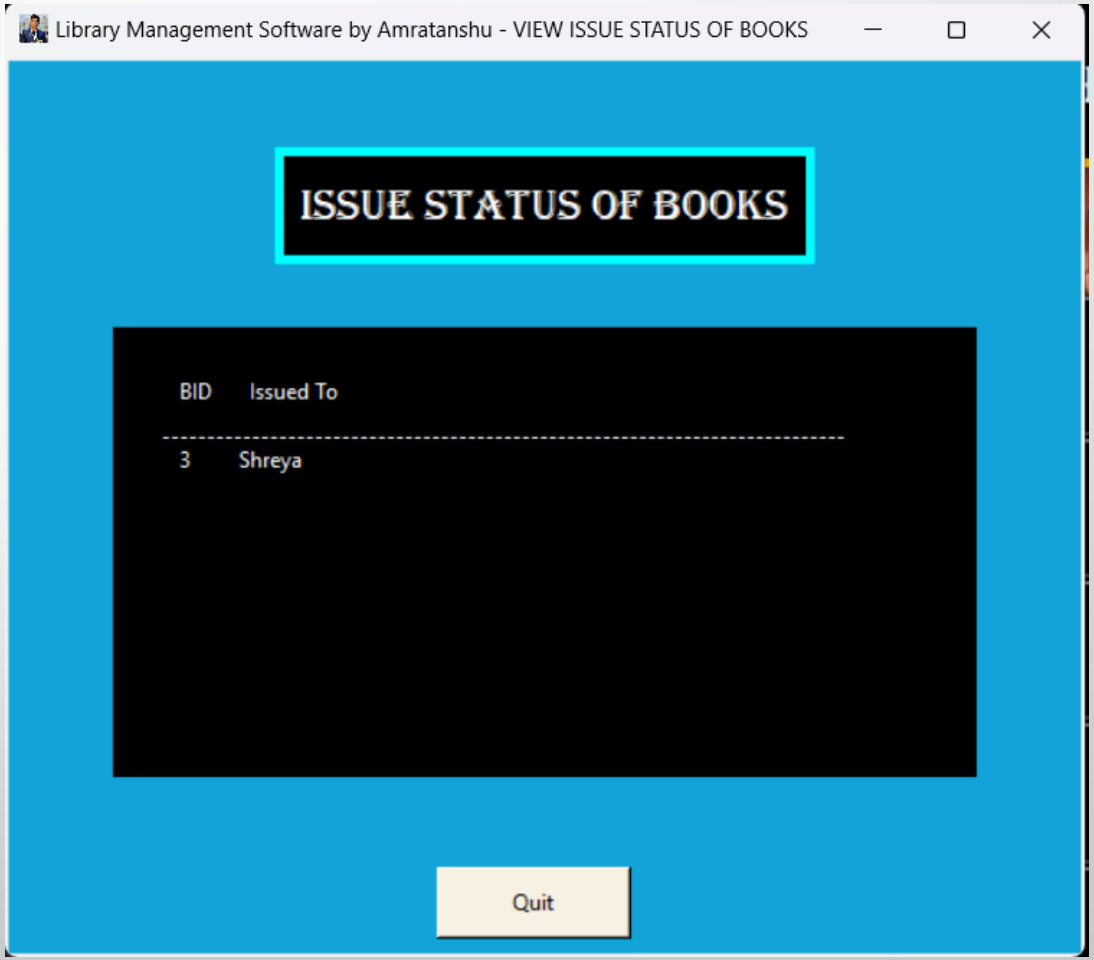
Module: ISSUE STATUS OF BOOKS

This module views a table of issued books from the database from the table **issued_books.**

It does not take any input from user and displays the list from the back-end.

The code for Return books is:

```
getBooks = "select * from
"+bookTable
```



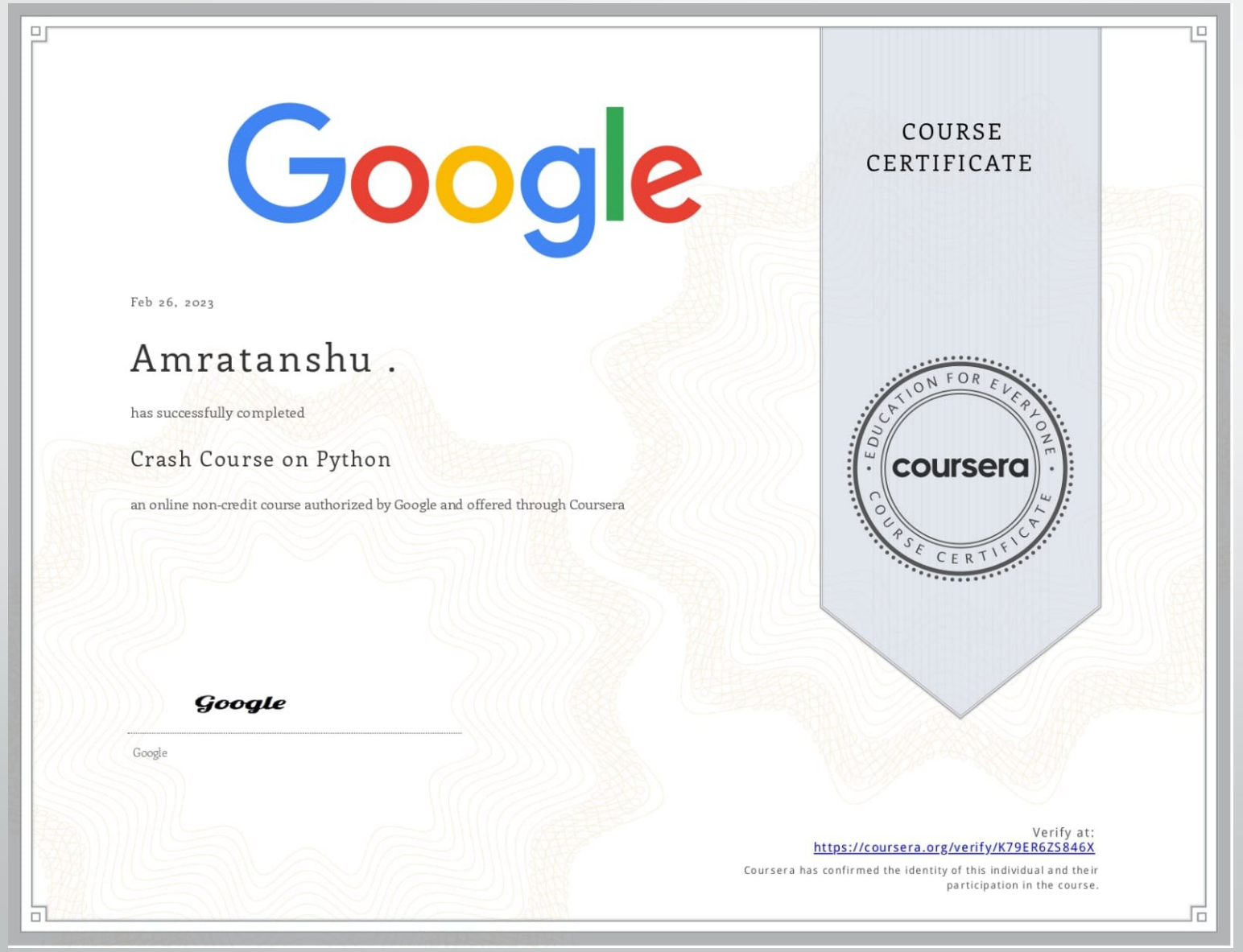
CONCLUSION



Library management system using python and tkinter with a decent UI. In order to make things easy, this tutorial divided the various tasks into different python files. In the real world, we follow such practices to make things easy to build. Hence, whenever making a project, divide your goals into modules and integrate them at a later stage.Apart from this, you can now take a step forward to extend the project by making a history tab which keeps track of the previous books issued. Moreover, you can integrate a login system to authenticate a user before making changes to the database.



CERTIFICATES



Crash Course On Python from Google
Date: 26-02-2023

CERTIFICATES



SQL Learning certificate from **SOLOLEARN**

Date: 25-02-2023

CERTIFICATES



Basics of Python Programming Certificate from **OpenWeaver**

Date: 21-10-2022


thank you